DBMS Model Practice Paper

Database Management Systems - Demo paper - 2025 edition - 50 marks

> *This is a self-created practice paper written for NoteNest. It is NOT an official university question paper. Use it to rehearse the style of exam questions.*

Section A - Short answers (2 x 5 marks)

Q1. Differentiate between a primary key, a candidate key and a super key with examples. **[5 marks]**

Q2. List the four ACID properties and explain Atomicity with a bank-transfer example. **[5 marks]**

Section B - Medium answers (3 x 8 marks)

Q3. Draw an ER diagram for a library (Member, Book, Loan) and map it to relational tables. **[8 marks]**

Q4. Write SQL to list departments whose average salary exceeds 50000, ordered by average salary descending. Explain why HAVING is needed. **[8 marks]**

Q5. Explain 2NF and 3NF. Normalize Order(order_id, product_id, product_name, qty) step by step. **[8 marks]**

Section C - Long answer (1 x 11 marks)

Q6. Explain the four isolation levels and the anomalies each one prevents. Give an example of a phantom read. **[11 marks]**

Answer hints

- Q1: primary key is the chosen candidate key; every candidate key is a minimal super key.
- Q4: use GROUP BY dept with HAVING AVG(salary) > 50000 because WHERE cannot filter aggregates.
- Q5: product_name depends only on product_id, a partial dependency, so split the table.